

West African Crime

Staffs Debs Stels

Introduction

This report is to present an analysis of crimes committed in Western Africa.



Figure 1: Bullet on the floor

To approach the analysis, we make use of base R functions and `dplyr` package for added analysis. The package is loaded using the `library()` function.

```
library(dplyr)
```

Firstly we create objects of the crime data

```
#|label: create-objects
```

```
country <- c(rep("Nigeria", 3), "Cameroon", rep("Nigeria", 3),  
  "Cameroon", rep("Nigeria", 8), "Cameroon", rep("Nigeria", 3))  
country
```

```
[1] "Nigeria" "Nigeria" "Nigeria" "Cameroon" "Nigeria" "Nigeria"  
[7] "Nigeria" "Cameroon" "Nigeria" "Nigeria" "Nigeria" "Nigeria"  
[13] "Nigeria" "Nigeria" "Nigeria" "Nigeria" "Cameroon" "Nigeria"  
[19] "Nigeria" "Nigeria"
```

```
length(country)
```

```
[1] 20
```

```
unique(country)
```

```
[1] "Nigeria" "Cameroon"
```

City preview

```
city <- c("Maiduguri", "Unknown", "Kano", "Bamenda", "Port Harcourt",  
  "Potiskum", "Damaturu", "Unknown", "Jos", "Lagos", "Gombe", "Kaduna",  
  "Logo district", "Konduga", "Warri", "Bama", "Buea", "Buni Yadi",  
  "Damboa", "Gamboru"  
  )  
city
```

```
[1] "Maiduguri" "Unknown" "Kano" "Bamenda"  
[5] "Port Harcourt" "Potiskum" "Damaturu" "Unknown"  
[9] "Jos" "Lagos" "Gombe" "Kaduna"  
[13] "Logo district" "Konduga" "Warri" "Bama"  
[17] "Buea" "Buni Yadi" "Damboa" "Gamboru"
```

This is a preview of the crime committed

```
n_crime <- c(488, 225, 130, 87, 76, 65, 63, 55, 51, 46, 42, 38, 37, 36, 35, 32,  
  31, 30, 30, 29)  
n_crime
```

```
[1] 488 225 130 87 76 65 63 55 51 46 42 38 37 36 35 32 31 30 30
[20] 29
```

Data Structure and type

To ensure proper analysis we have to investigate if the data are the right types and have the right structure.

```
typeof(country)
```

```
[1] "character"
```

```
typeof(city)
```

```
[1] "character"
```

```
typeof(n_crime)
```

```
[1] "double"
```

The data types are of expected, with `typeof()` exposing that `country` and `city`, are both character.

```
## Data Structure
str(country)
```

```
chr [1:20] "Nigeria" "Nigeria" "Nigeria" "Cameroon" "Nigeria" "Nigeria" ...
```

```
str(city)
```

```
chr [1:20] "Maiduguri" "Unknown" "Kano" "Bamenda" "Port Harcourt" ...
```

```
str(n_crime)
```

```
num [1:20] 488 225 130 87 76 65 63 55 51 46 ...
```

The number of elements in the object `country` is 20 and this is the same for `city` and `n_crime`.

Data analysis

The total number of crime committed is 1626. It would be nice to have indepth analysis about the countries and the crimes committed there. It is important to aggregate this data and change its structure to a `data.frame` to ensure smooth analysis.

```
crime_tbl <- data.frame(  
  country,  
  city,  
  n_crime  
)  
  
crime_tbl |> head()
```

	country	city	n_crime
1	Nigeria	Maiduguri	488
2	Nigeria	Unknown	225
3	Nigeria	Kano	130
4	Cameroon	Bamenda	87
5	Nigeria	Port Harcourt	76
6	Nigeria	Potiskum	65

Next, we separate regions according to the level of crime from high to low.

```
crime_tbl$crime_rate <- ifelse(  
  test = crime_tbl$n_crime >= 400,  
  yes = "High",  
  no = ifelse(  
    test = crime_tbl$n_crime >= 100,  
    yes = "Medium",  
    no = "Low"  
  )  
)  
  
crime_tbl |>  
  head() |>  
  knitr::kable()
```

country	city	n_crime	crime_rate
Nigeria	Maiduguri	488	High
Nigeria	Unknown	225	Medium
Nigeria	Kano	130	Medium

country	city	n_crime	crime_rate
Cameroon	Bamenda	87	Low
Nigeria	Port Harcourt	76	Low
Nigeria	Potiskum	65	Low

The total number of crime committed in each region is given below.

```
# Total number of crime and its average per country

aggregate(crime_tbl$n_crime, by = list(country = country), FUN = "sum")
```

```
  country    x
1 Cameroon 173
2  Nigeria 1453
```

Alternatively, we could this using `dplyr` and we can get the average crime per region in the process of getting the total.

```
crime_tbl |>
  summarize(
    .by = country,
    total_crime = sum(n_crime),
    average_number_of_crime = mean(n_crime)
  )|>
  knitr::kable()
```

country	total_crime	average_number_of_crime
Nigeria	1453	85.47059
Cameroon	173	57.66667

```
crime_tbl |>
  summarize(
    .by = country,
    total_crime = sum(n_crime),
    average_number_of_crime = mean(n_crime)
  )
```

	country	total_crime	average_number_of_crime
1	Nigeria	1453	85.47059
2	Cameroon	173	57.66667

The country with the highest crime is given below.

```
crime_tbl |>
  summarize(
    .by = country,
    total_crime = sum(n_crime)
  ) |>
  top_n(n = 1, wt = total_crime) |>
  knitr::kable()
```

country	total_crime
Nigeria	1453